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Introduction Chapter: Astronomy

Yann-Henri Chemin

1. Introduction

The observation of the Sun, the solar system, and the vast regions laying outside are commonly called *astronomy*.

In the whole range of observations of our neighborhood (**Figure 1**), we study the Sun, the Moon, the rocky planets, and the gas giants. In the last decades, the space probes sent around the solar system have enhanced our exploration capacity, that is, to go from hazy photographs to high-resolution mapping of most of our planetary bodies, even of their moons. Additionally, dwarf planets (Ceres and Vesta) and asteroids have been visited and mapped. This, in itself, it is a unique civilizational achievement in terms of exploration.

In between Mars and Jupiter, the asteroid belt is found. Further and mostly after Neptune’s orbit is the Kuiper belt. Eventually, the very possible Oort cloud, a reservoir of visiting comets, is vastly beyond the orbit of (90377) Sedna (**Figure 1**).

Our solar system is located in between two arms of a spiral galaxy, within what is often called a “finger” named the *Orion spur* (**Figure 2**). The center of our galaxy, the *Milky Way*, is the seat of a supermassive black hole (SMBH) called *Sagittarius A** [3].

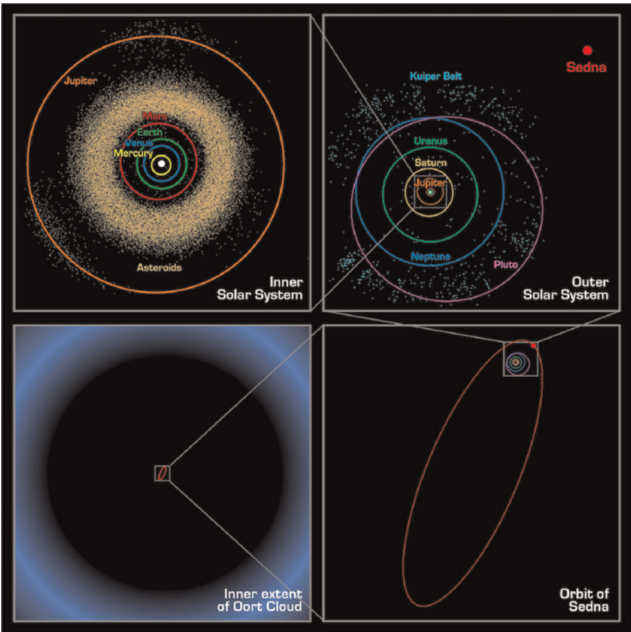


Figure 1. Extracted from ref. [1], this is the solar system. The upper left is defined by Jupiter’s orbit, upper right reaches Pluto’s orbit, the Kuiper belt, and the perigee of (90377) Sedna. Bottom right is the orbit of (90377) Sedna, which is barely seen in the bottom left alongside the inner Oort cloud, which is thought to be the source of comets. Image credit: NASA/CalTech.

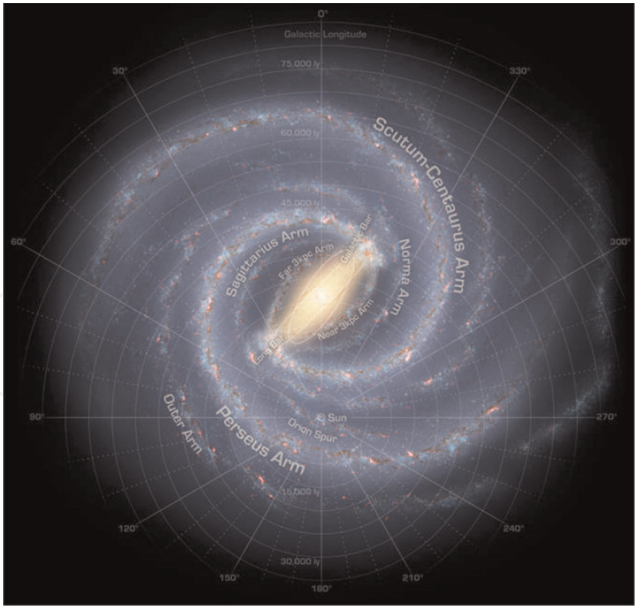


Figure 2.
The Milky Way. Extracted from ref. [2], image credit: NASA/Adler/U. Chicago/Wesleyan/JPL-Caltech.

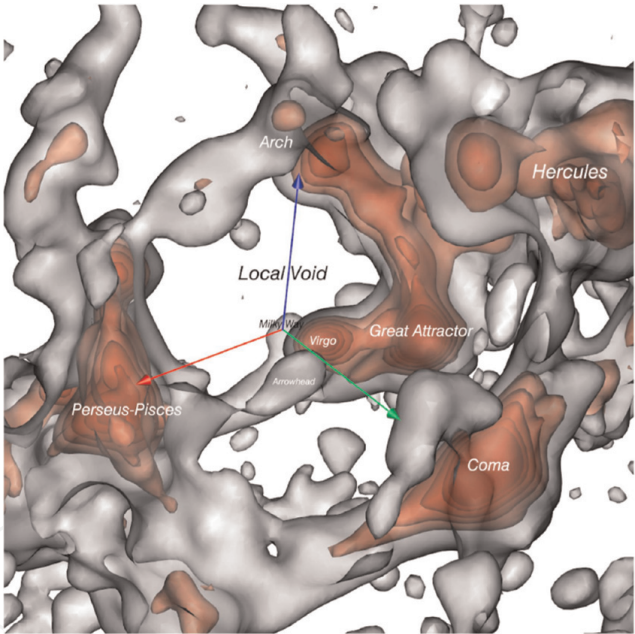


Figure 3.
Extracted from ref. [4], the local void, visualization centered on our galaxy, the milky way.

Our galaxy is located on a fringe of what could be called *mycelium* filaments. In other words, detectable matter at the cosmological level, tends to agglomerate in threads, interconnected by groups of larger material aggregations, not unlike the spread of fungal mycelium in Earth’s soil, punctuated by the presence of “nodes,” from which mycelium filaments extend. Of particular importance to our galaxy “suburb” is a large area void of matter (**Figure 3**). The *Local Void* has been mapped synthetically with a great resolution recently [4].

The overall observable universe is a sphere, centered on the location of the one observing. For us, it is planet Earth. When mapped, this sphere, so far, extends outwards in a radius of 46.5 billion light-years (440 Ym), as a comoving distance. The

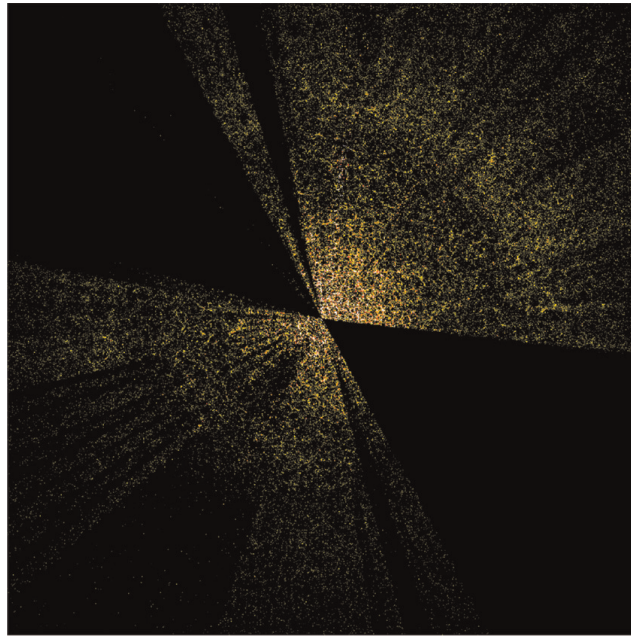


Figure 4.
 Extracted from ref. [6], scanning from the dark energy spectroscopic instrument (DESI), from the center, earth, toward the further seen so far, about one billion years after the big bang.

sloan digital sky survey (SDSS), mapping of the observable universe [5], was done not by direct distance, but in signed velocity from the observation point, deducting mostly from the red-shifting of the observations. The faster the positive red-shifting velocity, the further away the colocation to the observer (**Figure 4**).

2. Observing the sun and the stars

The closest of the stars, the Sun, is a G-IV, main-sequence star, and is located in the center of the Hertzsprung-Russell (HR) diagram [7] classification (**Figure 5**). Its next evolutions would be to leave the main sequence, that is the oblique line crossing **Figure 5** toward the upper right. It is then going to inflate and shift to become eventually a red giant, moving further up in the upper right arm of the HR diagram. Once reach maximum inflation, a cascade of gravitational collapses will happen, ejecting material by major explosions. Gravity will compact the remaining into a white dwarf, a neutron star, in the midst of its ejecta, witnessed as a (*super*) *nova* remnant, a nebula. The transition from the super-giant, the (*super*) *nova* reducing into the remaining white dwarf, will take the star rapidly across the HR diagram, from the upper right to the mid-upper left, and crossing down in the visible arc in the bottom left side of **Figure 5**.

The central part of the Sun is composed of a core (a fourth of its radius) where thermonuclear reactions generate energy. It has an average density ten times that of lead, and a temperature of 15×10^8 K. The radiative zone of the Sun is about one-third of the Sun's radius. Both transfer energy by radiative forces of photons. The pathways undergo a random walk and act as an apparent solid body [8].

The photosphere is 300–500 Km deep, and the light is emitted from there before the plasma becomes opaque. This is also the layer that defines the effective temperature of the Sun, about 5.8×10^3 K, plasma convection is visible there under the form of granules of sizes measured in Mm (10 m).

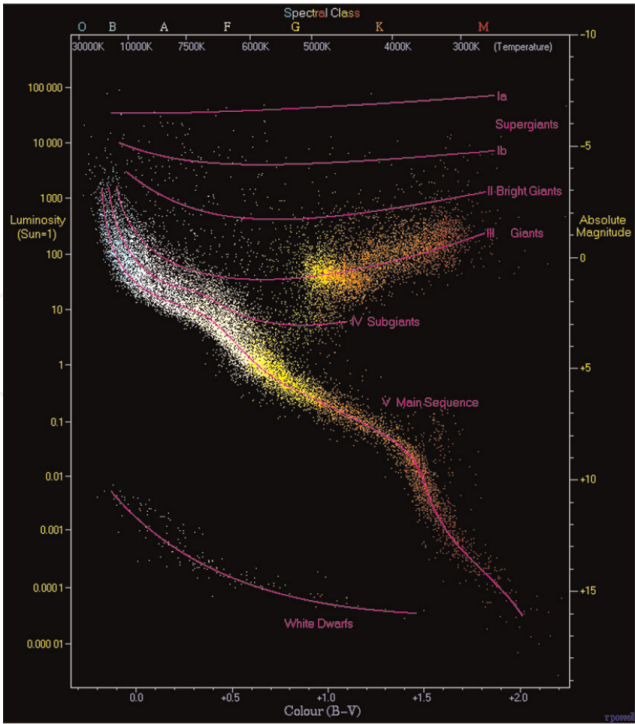


Figure 5.
HR diagram, classification of stars evolution. Image credits: Wikimedia.

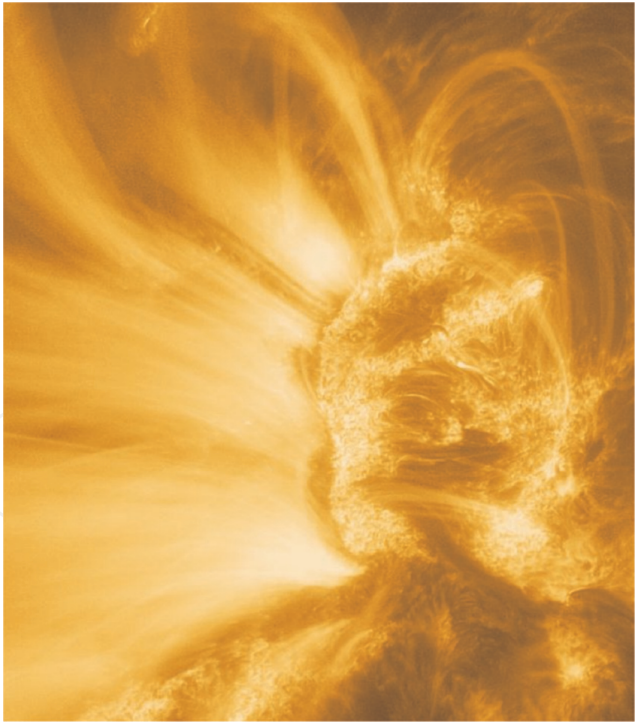


Figure 6.
High-resolution image of the sun from solar orbiter, showing magnetically bound plasma. Credit: ESA & NASA/ solar orbiter/EUI team; data processing: E. Kraaikamp (ROB). [9].

The Sun’s “atmosphere” (starting from **Figure 6**) is composed of the chromosphere and the corona (in that order). The chromosphere is 2000 Km deep and is the Sun’s eclipse “red ring of fire.” It has a steep drop in material density, and an initial temperature drop from $5.8 \cdot 10^3$ K to $3.5 \cdot 10^3$ K to eventually reach $35 \cdot 10^3$ K.

The corona is a very large volume above the chromosphere, vastly warmer too, made of ionized plasma of about 1106 K, with a majority of emission coming from Fe-XIV and Fe-X. It is the origin of the solar winds. Some areas with open magnetic fields yield faster solar winds (about $0.7 \cdot 10^6$ m/s).

We remotely sense the Sun (Solar Orbiter imagery in **Figure 6**), by analyzing electromagnetic spectra emitted from its activity. Thermodynamics and hydrodynamics applied to plasma with magnetic fields are all needed to study the radiative, convective, and exo-atmospheric conditions of the Sun energy transport.

More generally, stellar objects of different characteristics have long been observed and many physical theories have been developed relating observations and life cycles, that is, HR diagram in **Figure 5** and equations of stellar structure, respectively. Stellar oscillations [10], spherical harmonics, and resonance patterns analysis belong to geophysics and are now in common use to study and classify stars.

3. Observing the galaxies and the universe

In a similar way to stars, galaxies are also categorized along their paths of evolution. Hubble classification of galaxies evolution, the *Hubble Sequence* (**Figure 7**), reviewed here [11] (initial article [12]), provides an observation-based classification of galaxies.

The *Elliptical* galaxies start at spherical (E_0 ; $e = 0$) to the most common type of elliptical galaxies (E_7 ; $e = 0.7$). As the galaxy tends to age, central spin tends to send matter away in form of spiral arms.

The second level of classification in the *Hubble Sequence* (and further modifications) is dedicated to the extension and the shape of the *Spiral* arms of the galaxies. As seen in **Figure 7**, two branches of evolution differ in shapes of both the central bulge (whether it keeps spheroid or is barred) and the type of arms evolution. The first type, with spheroid central bulge is classified as Sa, Sb, and Sc along the evolution path. Similarly, *Barred* spiral galaxies are SBa, SBb, and SBc.

Sa (SBa) central bulge is bright and prominent.

Arms are tightly wound and smooth.

Sb (SBb) central bulge is less bright.

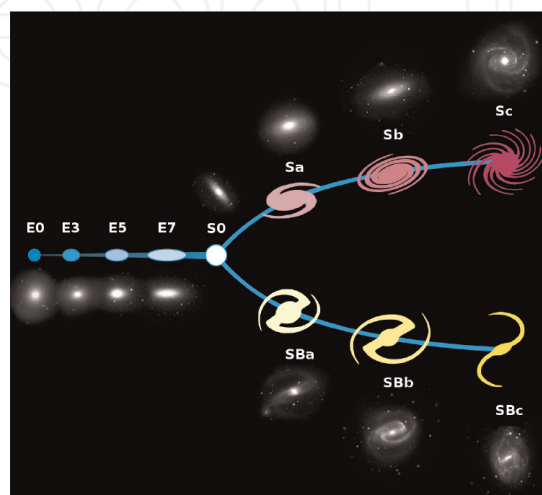


Figure 7.
 Hubble classification of galaxies evolution, the Hubble sequence. Courtesy: Wikipedia.

Arms are less tight than above.
Sc (SBc) central bulge is smaller and fainter.
Arms are loosely wound (stellar clusters and nebulae).
Sd (SBd) central bulge is dim.
Arms are bright and very loose, possible fragmentary arms.
The central bulge of our galaxy, the *Milky Way*, is dominated by a super massive black hole (SMBH), Sagittarius A* [13]. Sgr A*'s event horizon image is seen in **Figure 8**. It has an estimated mass of $4.152 \cdot 10^6 M_{\odot}$ and is the prime of several stars, their orbits helping define its mass. Its observed diameter is $51.8 \cdot 10^6$ Km, slightly more than the Sun-Mercury maximum distance ($\odot - \text{☿} = 46 \cdot 10^6$ Km at $\odot$ perihelion), which is about 1/3 AU (the mean distance $\odot - \oplus$).

Looking outside of the solar system has been largely enhanced with space telescopes. Furthering the capacity of the Hubble space telescope, in 2022, the James Webb Space Telescope (JWST) was activated at Lagrange 2, including the near-infrared spectrograph (NIRSpec) [14]. Its first images have been no less than revolutionary, giving direct observations of exoplanets and their atmosphere, but also looking further into the past of the universe.

The decades ahead of us promise the enhancement of our understanding of Sun, planets, the stars, black holes, and all other astronomical objects in our universe available to be observed. The observable universe itself just got smaller with JWST activated, and our understanding of the universe and its temporal unraveling is also furthering with every new data gathered. Time itself may also be better understood eventually, who knows?

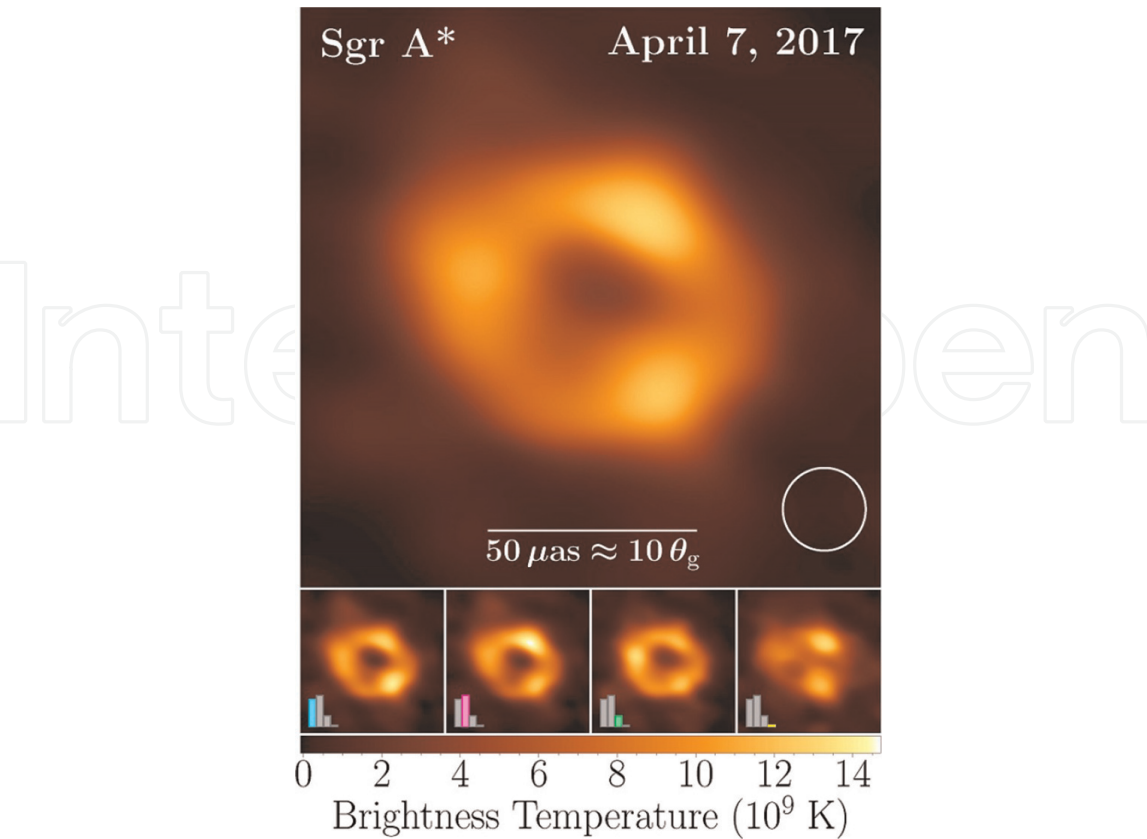


Figure 8.
The event horizon of the SMBH Sgr a* at the center of the milky way [13].

Abbreviations

☉	Sun
☿	planet Mercury
♁	planet Earth
AU	astronomical unit ($150 \cdot 10^6$ Km)
DESI	dark energy spectroscopic instrument
HR	Hertzprung-Russell
JWST	James Webb Space Telescope
NASA	US National Aeronautical and Space Administration
SDSS	sloan digital sky survey
SMBH	super massive black hole

Author details

Yann-Henri Chemin
European Commission, Joint Research Center, Ispra, Italy

*Address all correspondence to: dr.yann.chemin@gmail.com

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